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Ans ①:- operating system is software that helps user and hardware communicate. it controls CPU, memory, and I/O devices. main goals are making system easy and making performance better. OS gives interface, file system and process handling. It also provides security and controls resource use. It is important because without OS application cannot run properly.

Ans ②:- Process and thread are both execution units. Process is complete program with own memory. Thread is smaller part of process and shares memory with other threads. Thread operations are faster than process operations. Process switching is heavy because separate context, but threads are lighter and better for multitasking inside same application.

Ans ③:- CPU scheduling chooses which process gets CPU. works in arrival order and is easy to implement. But if ~~for~~ first process is large, others wait too much. Round Robin given fixed time slice to each ~~op~~ process and rotates. it is more fair and good for interactive systems. However it has ~~mon~~ more context swich switching, which can reduce efficiency if quantum is very small.

Ans ④ :- Deadlock is when two or more processes wait forever for resources. coffman conditions are mutual exclusion, hold and wait, no preemption, and circular wait. if these all happen deadlock can occur. Example: one process holds one lock and waits for second lock while another process holds second and waits for first.

Ans ⑤ :- virtual memory allows system to use disk as extra memory. Paging divides memory into pages and frames and maps them using page table. it allows non-contiguous allocation and supports larger ~~progr~~ programs. it reduces external fragmentation. But if too many page faults happen then system becomes slow.

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Ans (6) :- Synchronization is needed to avoid wrong results when many threads access same variable. critical section should be protected. mutex is lock/unlock mechanism for one thread at a time. Semaphore is counter based and can manage multiple resources also. These methods prevent race conditions and improve correctness.

Ans (7) :- Fragmentation is memory wastage. internal fragmentation is waste inside allocated block. External fragmentation is free memory split in many small blocks. internal can be reduced by better block sizing. External can be reduced by compaction and paging. paging is commonly used in modern operating systems.